Tutorial on Three phase Transformers and P.U

(1) Turns ratio per phase 10 ($N_1 : N_2$)

Supply voltage - 3300 V (line-line rms)

Load is adjusted such that it draws 10 A (line) currents from supply (3300 V side)

(a) Star - Delta

Star - HV side $\Rightarrow$ phase voltage $= \frac{\text{line voltage}}{\sqrt{3}}$ $I_{ph} = I_{line}$

Delta - LV side $\Rightarrow V_{ph} = V_{line}$ $I_{ph} = \frac{I_{line}}{\sqrt{3}}$

HV side phase voltage $= \frac{3300}{\sqrt{3}} = 1905.26$ V

HV side phase current = 10 A

LV side phase voltage $= \frac{1905.26}{10} = 190.526$ V

LV side phase current $= 10 \times 10 = 100$ A

$$\left[\frac{V_1 \text{ phase}}{N_1} = \frac{V_2 \text{ phase}}{N_2} \qquad I_1 \text{ phase } N_1 = I_2 \text{ phase } N_2 \right]$$

LV side line voltage = 190.526 V (∵ Delta connection)

LV side line current $= 100 \times \sqrt{3} = 173.21$ A

$KVA = \sqrt{3} V_L I_L = \sqrt{3} \times 190.526 \times 173.21 = 57.15$ kVA

(b) $\Delta - Y$

LV side phase voltage $= \frac{3300}{10} = 330\ V$

LV side line voltage $= \sqrt{3} \times 330 = 571.576\ V$

LV side phase current $= \frac{10}{\sqrt{3}} \times 10 = 57.735\ A$

LV side line current $= 57.73\ A$

Output KVA $= \sqrt{3} \times 571.576 \times 57.73 = 57.16\ KVA$

(2) 100 MVA, 230 / 115 KV $\Delta - \Delta$

R p.u $= 0.02$ p.u

X p.u $= 0.055$ p.u

load is inductive of 80 MVA at pf $- 0.85$ lag

Voltage at load $- 115 \times 10^3$ V

Given load MVA

$$\sqrt{3} V_L I_L = 80 \times 10^6$$

$$I_L = \frac{80 \times 10^6}{\sqrt{3} \times 115 \times 10^3} = 401.635\ A$$

$$\sqrt{3} V_{L\,base} I_{L\,base} = 100 \times 10^6$$

$$I_{L\,base} = \frac{100 \times 10^6}{\sqrt{3} \times 115 \times 10^3} = 502.04\ A$$

$$|I_L\ p.u| = \frac{401.635}{502.04} = 0.8$$

$$\phi = -\cos^{-1}(0.85) = -31.788^\circ$$

$$I_L\ p.u = 0.8 \angle -31.788^\circ$$

$$V_1 = V_2 + I_2 Z = 1\angle 0^\circ + 0.8\angle -31.788^\circ \times (0.02 + j0.055)$$

$$= 1.03718\angle 1.6^\circ$$

$$\text{Voltage regulation} = \frac{1.03718 - 1.0}{1.0} = 3.718\ \%$$

(3) 150 KVA, 2400/460V

$Z = 0.038 + j0.135$ pu on its own base

load = 95 kVA at UPF at 438 V

Voltage on HV side

Base voltage = 460 V

Base KVA = 100

Z on 100 KVA base

$$Z_{pu} = \frac{0.038 + j0.135 \times 100}{150} = 0.02533 + j0.09\ p.u$$

$$V_{pu} = \frac{438}{460} = 0.95217\ p.u$$

$$I_2 \text{ actual} = \frac{95 \times 10^3}{\sqrt{3} \times 438} = 125.224\angle 0^\circ \quad (UPF)$$

$$I \text{ base} = \frac{100 \times 10^3}{\sqrt{3} \times 438} = 125.5109$$

$$I_{pu} = \frac{125.224\angle 0^\circ}{125.5109} = 0.9977\angle 0^\circ\ pu$$

$$V_{1pu} = V_2 + I_2 Z = 0.95217 + 0.9977(0.02533 + j0.09)$$

$$= 0.981557\angle 5.248^\circ\ p.u$$

Actual Voltage on HV side = $V_{1pu} \times V_{base}$ $= 0.981557 \times 2400$

$= 2355.74$ V

(4) A $\Delta\text{-}\Delta$ transformer is made with three identical single phase transformers of ratings

100 kVA 2400/120V, 50 Hz

feeder impedance $-0.3+j0.8\ \Omega$

Voltage at sending end of feeder 2400 V line-line

From 1-ϕ sc test LV side shorted

$$Z_H = \frac{V_H}{I_H} = 1.382\ \Omega$$

$$R_H = \frac{P}{I_H^2} = 0.5056\ \Omega$$

$$X_H = \sqrt{Z_H^2 - R_H^2} = 1.286\ \Omega$$

transformer per phase impedance referred to HV side

$$Z_H = 0.5056 + j1.286$$

(a) Secondary (LV) line-line voltage

rated current = ?

rated power = 100x3

rated voltage line-line (LV) = $120\times\sqrt{3}$

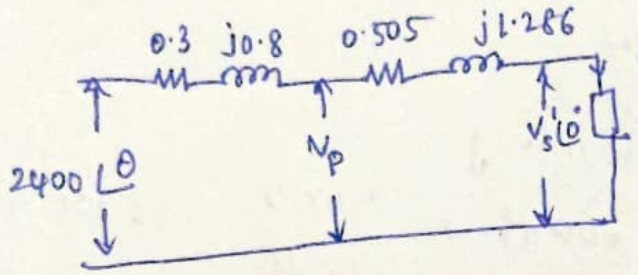
$$\text{rated current} = \frac{P_0}{\sqrt{3}\times V_L} = \frac{100\times3}{\sqrt{3}\times\sqrt{3}\times120} = 833.34\ A$$

load current = $833.34\angle -0^\circ$ A UPF

primary phase current = $\frac{833.34\times120}{2400}\angle 0^\circ$

$= 41.667\angle -0^\circ$ A

equivalent circuit referred to HV



$$2400\angle\theta = V_s'\angle 0° + 41.667\angle 0°\,(0.3+0.505+j(0.8+1.286))$$

Taking magnitudes

$$(2400)^2 = (V_s' + 41.667\times 0.805)^2 + (41.667\times 2.086)^2$$

$$\Rightarrow V_s' = 2365.1013\ V$$

referred to LV side

$$V_{s\,ph} = \frac{2365.1013\times 120}{2400} = 118.255\ V$$

$$V_{s\ line-line} = 118.255\times\sqrt{3} = 204.8237\ V$$

(b) under S.C in secondary terminals

$$\text{current} = \frac{2400}{Z_t+Z_t} = \frac{2400}{0.805+j2.086} = 1076.0884\ A$$

$$\text{HV winding current} = \frac{1076.0884}{\sqrt{3}} = 621.279\ A$$

current per phase is 1076.0884 but current flowing through Δ winding is 621.279 A

$$\text{LV winding current} = 621.279\times\frac{2400}{120}$$

$$= 12.425\ kA$$

(5) 3, 1-ϕ 25 kVA, 3300/300 V Δ-Δ

load power = 50 kVA

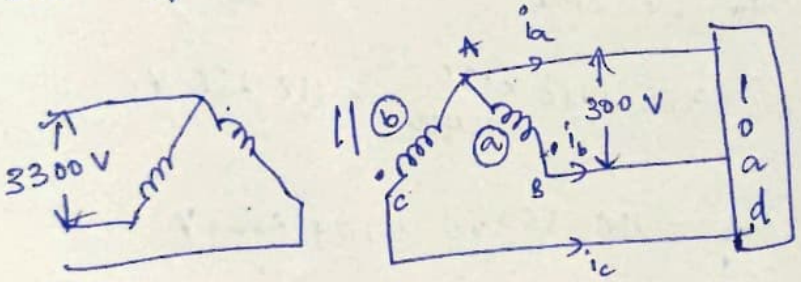
(a) current in 1° winding

Secondary line current = $\frac{50\times10^3}{\sqrt{3}\times300} = 96.225$ A

$\Rightarrow$ 2° phase current = $\frac{96.225}{\sqrt{3}} = 55.55$ A

$\Rightarrow$ 1° phase current = $55.55 \times \frac{300}{3300} = 5.05$ A

In open Delta configuration



(a) KVA supplied by each transformer =

Secondary phase current × phase voltage (= line vol)

In open delta case 2° phase current = 1° line current

KVA by each tf = $96.225 \times 300 = 28.8675$ kVA

To find kVA rating we should know the maximum current tf can supply

I_{max} (line) = I_{max} phase = $\frac{25\times10^3}{300} = 83.33$ A

kVA rating = $\sqrt{3}\, V_L I_L = \sqrt{3}\times300\times83.33$

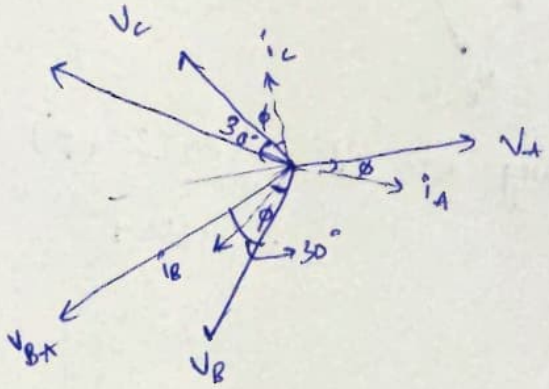
$= 43.301$ kVA.

(b) If the load draws current at lagging pf of 0.95 power factor of operation of each transformer the current flowing out terminal is consider positive (given • in circuit)

now voltage across winding (a) is V_{BA} and current is i_B

Voltage across winding (b) is V_{CA} and current is i_C

phasor is given below to find the angles



$\phi = \cos^{-1}(0.95) = 18.194$

For transformer winding (a) $Pf = \cos(30 - 18.194)$

$= 0.9778$ Lead

For transformer winding (b) $Pf = \cos(30 + 18.194)$

$= 0.67$ lag